

Lesson 4.2 Choose our Audience and Plan our Approach

Location: Classroom

Grade Level: High school

Time: 50 minutes

Driving Question

- How can we select a science communication approach for the right audience?

Lesson Synopsis

In this lesson, students continue working on their science communication products. They will **build the profiles of their target audience** and **choose their communication approach**. By the end of the lesson, they should have a preliminary science communication plan and a defined target audience to test in Module 4.3.

What students will figure out:

- Human activities contribute to climate change in ways that affect different groups differently (e.g., urban vs. rural, youth vs. policymakers, industries vs. communities).
- Audience matters in our science communication. This is because the impacts of climate change are uneven, people have diverse priorities, and thus we should tailor communication strategies to these audiences.
- Communication choices and media can shape our audience's reaction. For example, the way data are represented to emphasize risks versus solutions can influence how audiences perceive urgency and possibility for change.

LEARNING OUTCOMES

By the end of this investigation, students will be able to...	You can assess this using...
1. Identify the target audience of their science communication products.	Student discussion and science journal in the Investigate section.
2. Create a preliminary plan for the science communication products, including the product type (e.g., podcast, social media post, game)	Student science communication plan draft in the Investigate section.

Building towards NGSS and AI Literacy:

- HS-ESS2-4:** Use a model to describe how variations in the flow of energy into and out of Earth's systems result in changes in climate.

LESSON OVERVIEW

Section	Time	What Happens
Navigate	10 minutes	Students get introduced to the lesson goals: identify key audiences and plan their science communication product.
Investigate	30 minutes	Students work on their science communication product plan.
Navigate	10 minutes	Students share their plans and get peer feedback.

MATERIALS

- [Lesson 4.2 slides](#)
- [Science journal](#)
- [Teacher Rubric](#) for science communication products (referenced throughout module 4)
 - [Student Rubric Checklist](#) (provided in science journal, p.60)

BEFORE YOU START TEACHING

1. Review the [Student Rubric Checklist](#) (also in Science Journal p. 60) and [Teacher Rubric](#) to know the expectations of students.
2. Throughout the program, students will be working in teams of three (preferably).
3. Make sure that you have a way to project a video with sound.
4. The science journal is intended to be a digital notebook. However, if you decide to print copies for students, they may need a writing instrument (pen, pencil, etc.).

WHERE WE ARE GOING AND NOT GOING

Where We Are Going: Students will deepen their science communication skills by focusing on the **audience**—who they are, what they care about, and how they receive information. They will explore different communication approaches (social media, podcast, flyer/infographic, presentation, game) and choose the one that fits their audience best.

Where We Are NOT Going: Students are not yet creating or finalizing their science communication products (no polished Instagram posts, finished flyers, or recorded podcasts today). They will mostly be building on the evidence that they have already modeled and can conduct additional information search if needed. It is important to highlight in facilitation that students are not trying to persuade every possible audience. Instead, they will be focusing on one target audience and planning for them.

LEARNING SEQUENCE

Lesson plan notes are also provided on the slide notes.

NAVIGATE: Introduction to the Lesson (10 minutes)

1. Open the slideshow to **Slide 1** and introduce the lesson.
2. Advance to **Slide 2**. Inform students that we will use the science communication principles from the previous lesson (lesson 4.1) to develop their plan and that today, they will work on planning their message further by choosing their audience and planning their approach.
3. Advance to **Slide 3**. Share the **Central Inquiry Question**: *What makes a science communication message clear, accurate, and engaging for a target audience?*
4. Advance the slideshow to **Slide 4**. Share that inclusive and effective science communication often involves five key areas: clarity and language, content, knowledge organization, style, and audience engagement.
 - a. For this lesson, we are focusing on building on (1) **clarity and language** and (2) **content** from the previous lesson. We will also include **audience engagement**.
 - i. Recall how **clarity and language** help us deliver often complex information so that the public or our target audience can understand. This may often include translating or including language for diverse populations.
 - ii. Recall how **content** ensures that the information is not only accurate but also has components of connection to everyday life, or even your personal/community life.
 - iii. Introduce how audience engagement is focused not only on inspiring reflection and action through tailored messaging but also on inviting a conversation with the audience.
5. Advance the slideshow to **Slide 5**. Share how this is an example of a science communication product created by high school students.
 - a. Ask: *What is something you notice and wonder about?*
6. Advance the slideshow to **Slide 6**. Share that in this example, students created a science communication product in the form of a Lotería game. It is a traditional Mexican game similar to Bingo that uses illustrated cards representing culture, history, daily life, and folklore to bridge scientific content to students' cultural experiences growing up playing the game.
7. On **Slide 7**, reflect with students on the **Central Inquiry Question**: *How can we make a science communication message clear, accurate, and engaging for a target audience?* Take a closer look at the science communication product example's message, scientific accuracy, and real-world and personal connection.

- a. **Language and clarity:** *Is the information accessible to the target audience? Is it translated to meet the language needs of the intended audience (optional)?*
 - b. **Audience Engagement:** *Does the game fit the audience of middle school students (target audience)? Is there a clear call to action or prompt for reflection?*
8. On **Slide 8**, discuss: *What works well? And what can be improved?*
9. Advance the slideshow to **Slide 9**. Remind students that, based on the five components of inclusive and effective science communication, a **student checklist** has been created to guide them in reviewing and designing science communication products.
10. Advance to **Slide 10** and share that our main focus is revising our messages to fit the language and engagement needs of our target audience. Therefore, we will specifically look at the (1) **language and clarity** and (2) **audience engagement** parts of the checklist.
11. Advance to **Slide 11**. Explain to students that the following slides will show five different possible products that students could create. This time, we are not only focused on their message but also how they tailor the product to their specific audience.
 - a. Instruct students to use the Student Science Communication Product checklist on p. 60 in the Science Journal to review 2-3 different science communication products (Slides 12-16)
12. Advance to **slides 2-6** that provide examples of science communication campaigns, the target audience of these campaigns, and the steps that the campaigns take. The examples cover the science communication options students will work on: *infographics, flyers, podcasts, social media posts, and board games*.

Additional guiding prompts: *What are some identifying features of these science communication examples? How might these products cater to different audiences? What might you borrow for your own communication plan?*

Possible student ideas:

- Social media posts (Instagram/TikTok): the target audience could be teens and young adults. Key features: short duration, hashtags, funny/colorful content, emotional appeal.
 - Infographic/flyer: the target audience could be the general public. Key features: visuals, simple text, key facts.
 - Townhall presentation: the target audience could be decision-makers. Key features: data incorporation (e.g., graphs), call to action, persuasive language.
 - Board games: the target audience could be a youth or educational audience. Key features: sets of game rules, interactions, storytelling/scenario elements.
13. Advance to **Slide 17** for a whole-class discussion.
 - a. Sample guiding questions:

- i. *As you look across all four products, who was the intended target audience? How was the product's message tailored for the audience? What were some ways that they engaged the audience?*
- ii. *Pick a product that may struggle to connect to its audience. What is one specific revision you would suggest to improve its messaging?*

INVESTIGATE: Plan the Science Communication Plan (30 minutes)

1. Advance to **slide 18** and share with students that their next step is to build a profile for who they want their audience to be. Ask: *Who is your target audience? What is their background? What motivates them or concerns them the most? What are their preferred communication strategies (how do we reach this audience)?*

Explain to students that one way to sketch out these audience profiles is to think of a fictional “persona” and create an audience card that describes their key characteristics. The slide displays examples of the audience cards that identify the key audience and their priorities.

2. Advance to **slide 19**. In their project teams, students will start planning their science communication approach. They will draft their plan in response to the following questions (Students may use [Science Journal pages 66](#)):
 - a. *Recall the message that you drafted in lesson 4.1. What science communication option will you pick to best convey your message? For example, will it be a social media post, flyer, infographic, podcast, presentation at a city council meeting, or a game?*
 - b. Ask students: *Why is this product effective for them?*
 - c. Have students brainstorm in their project teams: *What does the product include: text, visuals, data, stories, or calls to action?*
3. Advance to **slide 20**. Show students an example of mapping an example science communication product (from **slides 2-3**) to components of the graphic organizer (see below).
4. Advance to **slide 21**. Tell students they can use a graphic organizer to brainstorm and sort ideas for their science communication plans.

Target audience	
Key message	
Text	
Visual	
Data from our model	
Emotional appeal	

5. Remind students that they are not yet building the products and are just making plans for what the products might include. The outcome at the end of the lesson should be a **filled-out table** (i.e., the graphic organizer) of their science communication plan draft.

NAVIGATE: Share our Plan (10 minutes)

1. Advance to **slide 22**. Invite student teams to share their initial plans. Invite others to provide feedback about what they like and suggestions to improve the plans.

Student teams can use the format: *"Our audience is _____. We will use a [product type] because _____. Our key message is _____, and we'll use [this data] to support it. We are thinking of using [this visual] and [this text]."*

The educator can highlight to students how their message shifts, depending on who they are talking to. That's what real science communicators do.

2. Optional: [Student Reflection in Science Journal p. 68](#)